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STOCHASTIC PROGRAMMING

PROBLEMS SESSION 1

6. A company wants to invest an amount of money in $n$ financial assets (shares, bonds, etc.). It wants to decide which fraction to invest in each asset $i = 1 \ldots n$. Let $x_i$ be the fraction invested in asset $i$. Acquiring assets has expenses of $f_i\%$ for each asset $i$. The future returns are given by the random variables $\xi_i$ for each asset, which are assumed to be independent. Formulate the two-stage stochastic optimization problem to be solved by the company. Simplify as much as possible the second stage, obtaining the deterministic equivalent formulation.

We want to minimize investment costs while having the maximum possible return

$$\min_{x_i} \sum_{i=1}^{n} f_i \cdot x_i + \mathbb{E}[Q(x, \xi(\omega))]$$

$$\sum_{i}^{n} x_i = 1$$

$$x_i \geq 0 \quad \forall i \in \{1, \ldots, n\}$$

Q corresponds to the problem of obtaining maximum return

$$Q(x, \xi(\omega)) = \min_{y_i} \sum_{i=1}^{n} -\xi_i(\omega) \cdot y_i(\omega)$$

$$y_i(\omega) = x_i \quad \forall i$$

$$y_i(\omega) \geq 0 \quad \forall i$$

Now, $\mathbb{E}_\xi[Q(x, \xi)] = \mathbb{E}\left[\sum_{i=1}^{n} -\xi_i(\omega) \cdot x_i\right] = -\sum_{i=1}^{n} x_i \, \mathbb{E}[\xi_i]$

So, if we plug this in the original problem



$$\min_{x_i} \sum_{i=1}^{m} \left[ f_i - \mathbb{E}(\xi_i) \right] x_i$$

$$\text{s.t.} \quad \sum_{i=1}^{m} x_i = 1$$

$$x_i \geq 0 \quad \forall i$$

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